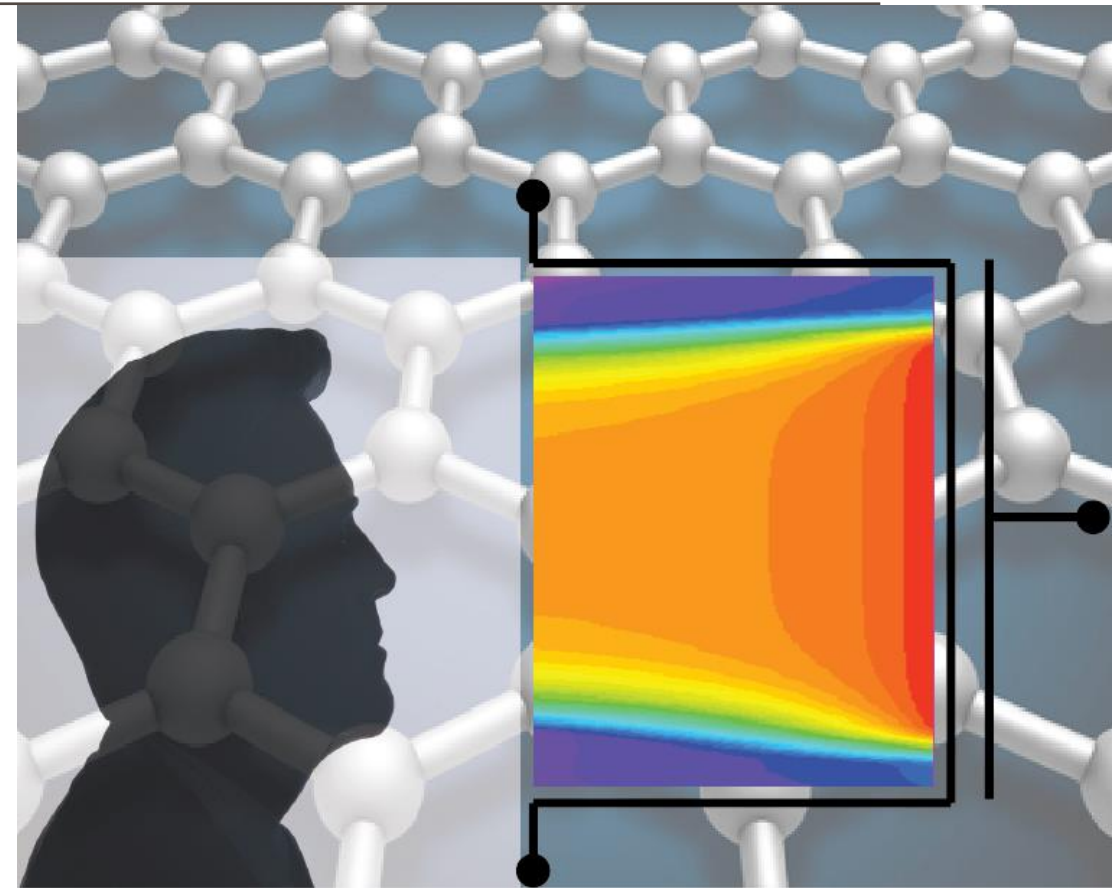


ADVANCED SOLID-STATE DEVICES

Week-1, Lecture-1

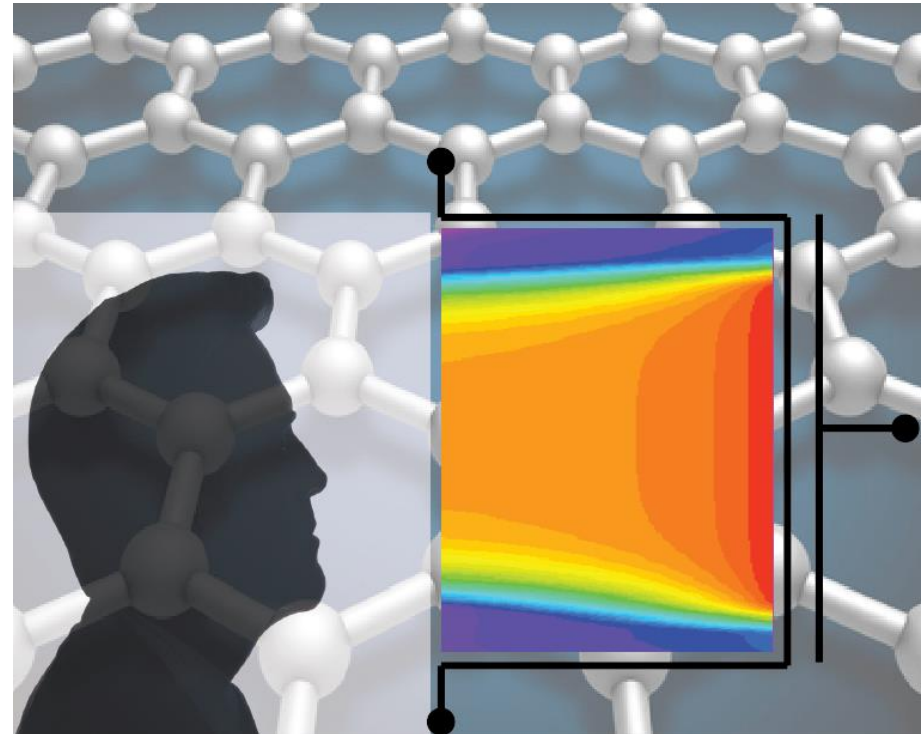
Sneh Saurabh
9th January, 2025



About the Course

Advanced Solid-state Devices (ASD): Objective

- Understand the fundamental principles and advanced concepts related to MOSFETs and BJT
- Familiarize with futuristic solid-state devices that will help undertake device-centric research



ASD : Expected outcome

- Clearly explains operating principle and terminal characteristics of diodes, MOSFET and BJT, including non-ideal effects.
- Is able to compute a given parameter or physical quantity for PN junction, MOSFETs, and BJT, including non-ideal effects.
- Is able to evaluate the challenges of scaling traditional MOSFETs and justify the need for advanced solid-state devices.
- Is able to analyze the operating principles of some of the advanced solid-state devices and judge their merits, demerits and challenges.



ASD: Course Content

Week No.	Topic
1,2	Basics of semiconductor physics: density of state, Fermi distribution function, semiconductor in equilibrium, dispersion relation; effective mass; carrier transport, recombination, PN junctions: ideal behavior and non-ideal effects, capacitances, small signal equivalent circuit, breakdown; Metal-semiconductor junction: barrier height, rectifying and ohmic, Schottky barrier lowering, thermionic emission and tunneling
3-6	Field-Effect; Field Effect Transistors: JFET, MESFET, and MOSFET; MOS Capacitor: fundamentals, band diagram, surface potential, inversion, flat band voltage, threshold voltage and its modelling; Parasitic charges in MOSFET: types and effects; CV measurements and frequency effects; body effect; MOSFET operation: input and output characteristics, saturation, transconductance, small signal model and frequency limitations; Non-ideal effects in MOSFETs: channel length modulation, velocity saturation, subthreshold current, DIBL, hot electron effects; MOSFET CAD SPICE models (BSIM);
7-8	BJT: structure, biasing, carrier profile, transistor action, current gain; BJT: Non-ideal effects: base width modulation, high-level injection, emitter bandgap narrowing, current crowding, breakdown; hybrid-pi model and frequency limitations;
9	Introduction to device simulation, circuit simulation, and compact models;
10-13	Scaling and limitations of traditional MOSFETs; Nano transistors: ballistic and diffusive transport; Advanced MOSFETs: high-k gate dielectric, strained silicon and choice of channel material; SOI MOSFETs: PDSOI and FDSOI; FinFET; GAA FET; Stacked nanosheet FET; Tunnel FET; Negative Capacitance FET; IRDS and futuristic devices; Presentations

Who can/cannot take ASD course?

- B.Tech. students: can take only if PSD (Physics of Semiconductor Devices) taken (because physics of semiconductors will be assumed to be known)
- BTech/MTech/PhD students: cannot take if SSD (Solid-state Devices) course already done in previous semesters (because more than 50% course content is same as SSD)
- Expected that you know basic concepts related to physics of semiconductors

ASD: Schedule, Office Hours, TAs

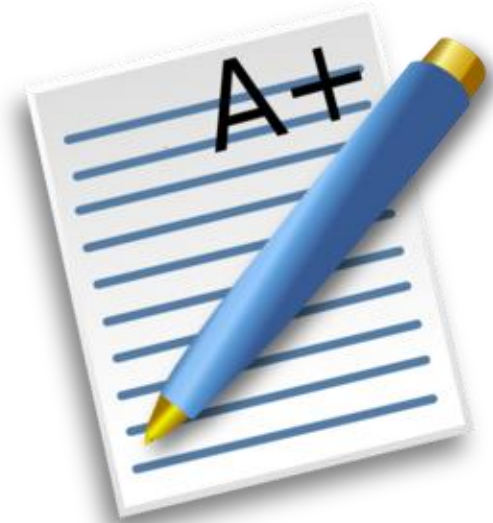
- Lecture: A106
 - Monday and Thursday: 9:30 pm - 11:00 pm
- Instructor's Office Hours:
 - Monday, Thursday: 11:30 pm - 12:00 pm
 - Wednesday 3:00 pm - 4:00 pm

- TAs
 - Varun
 - Anshu Kumar

- Join Google Classroom:
 - Code: **bx2ewm7**

ASD: Evaluation Criteria

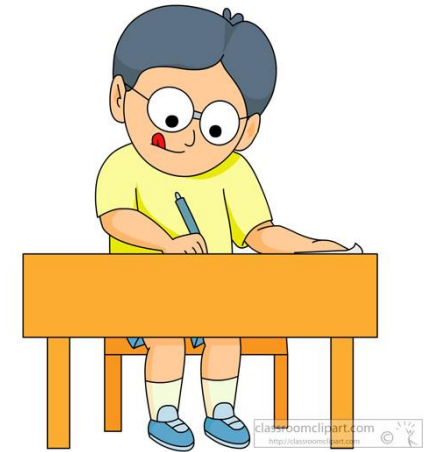
Type of Evaluation	ECE613
Assignment	15 %
Presentation on Advanced Topic	15 %
Mid-semester Exam	30 %
End-semester Exam	40 %



- Less than 30% in absolute term: FAIL grade

ASD: Assignment (1)

- 2 Assignments to be done individually
- Final score will be scaled to 15%
- Answer should be hand-written on A4 size paper and submitted in Google classroom (no printout submission allowed).



	Assignment 1 (Tentative Dates)	Assignment 2 (Tentative Dates)
Assignment Upload	15 th February	15 th March
Time given	1 Week	1 Week

ASD: Assignment (2)

- Be punctual in online submission
 - Each day of delay will attract 20 % penalty and after 5 days delay no more submission
 - No excuse of internet down or only a few seconds of delay (Try to upload well in advance)

Presentation

- To be done in a group of three
 - You need to select a topic and get it approved
 - Google sheet to be floated for entering your topic and getting it approved
 - Presentation schedule will be announced later (may be on weekends)
- The topics should be on Solid-state Devices related to one of the following areas:
 - Neuromorphic Computing
 - Stochastic Computing
 - Probabilistic Computing
 - Brain-inspired Computing
 - Quantum Computing
 - Post-CMOS Computing
 - Any other advanced computing with instructor's approval

Mid-semester/End-Semester Exam

- Closed book
- Need not memorize formulae
- **One A4 sheet** with formulae written in your handwriting will be allowed during exam

ASD: Reference Books

D. A. Neaman, "Semiconductor Physics and Devices"
Yannis Tsividis, "Operation and Modeling of the MOS Transistor"
Y. Taur and T.H. Ning, "Fundamentals of Modern VLSI Devices"
S. M. Sze & M. K. Lee, Semiconductor Devices, Physics and Technology

Policies

ASD: Attendance Policy

- Student with less than 30% attendance [less than 8 classes] will be awarded FAIL grade *at the discretion of the instructor* [even when the total marks is above the PASS grade]

ATTENTION: DO NOT Register for the course if you do not plan to attend the classes regularly (for ANY reason!)

ASD: Policies (1)

- Be punctual to class: up to 5 minutes late is tolerated, beyond that please do not come
 - Bring calculator everyday
-
- Cell phones, Laptops, IPODs, MP3 players or other portable electronic devices of any kind NOT allowed in classroom (should be switched off)



ASD: Policies (2)

- Academic dishonesty/plagiarism/cheating
 - IIITD rules will apply
 - Academic dishonesty policies of IIT Delhi apply.
<https://www.iiitd.ac.in/sites/default/files/docs/education/AcademicDishonesty.pdf>
 - Checks will be in place to detect copying/plagiarism/cheating



ASD: Questions/Doubts

1. Class: Many may have the same questions
2. Just After Class
3. Office Hours
4. During revision
5. No last minute query for assignment submission
 - Clear all your doubts at least 2 days before submission



ASD: Expectation

- Revise last day lecture before coming
 - Revision will be done at the start of every lecture
 - Students participation will be ensured
- Ask questions and be active participants
- Course expected to be at pace so that **EVERYONE** is able to follow
- Give feedback

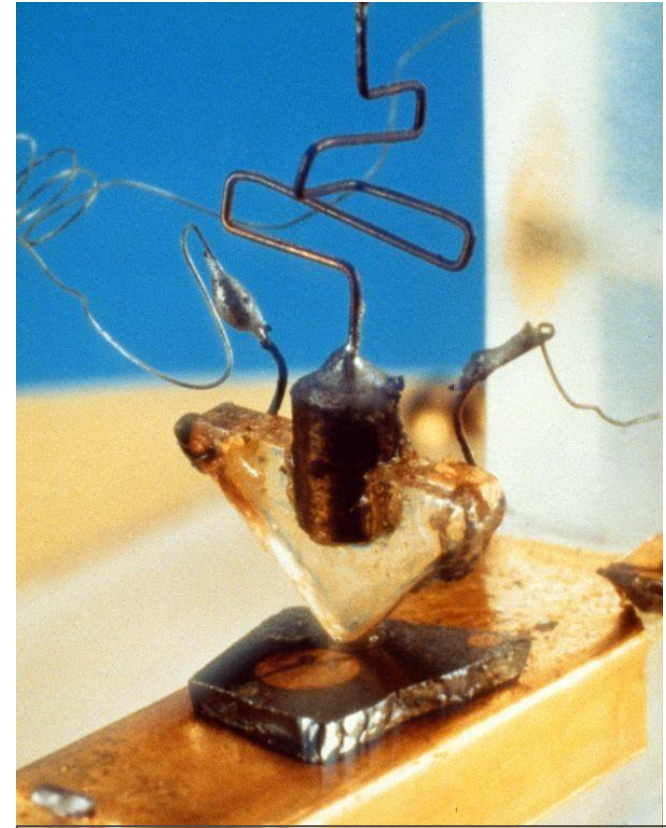


Solid State Devices: Overview

Solid-state electronics

What is ***solid-state*** electronics?

- Electronic action of devices occurred in a solid-state (as distinct from vacuum tubes)
- No movable parts (SSD memory in contrast to HDD)
- Practically, it means **semiconductor**-based electronics
- Took off very fast after the invention of semiconductor-based transistor



Source:
<https://www.computerhistory.org/revolution/digital-logic/12/273>

Semiconductors

Table 1.1 | A portion of the periodic table

III	IV	V
5 B Boron	6 C Carbon	
13 Al Aluminum	14 Si Silicon	15 P Phosphorus
31 Ga Gallium	32 Ge Germanium	33 As Arsenic
49 In Indium		51 Sb Antimony

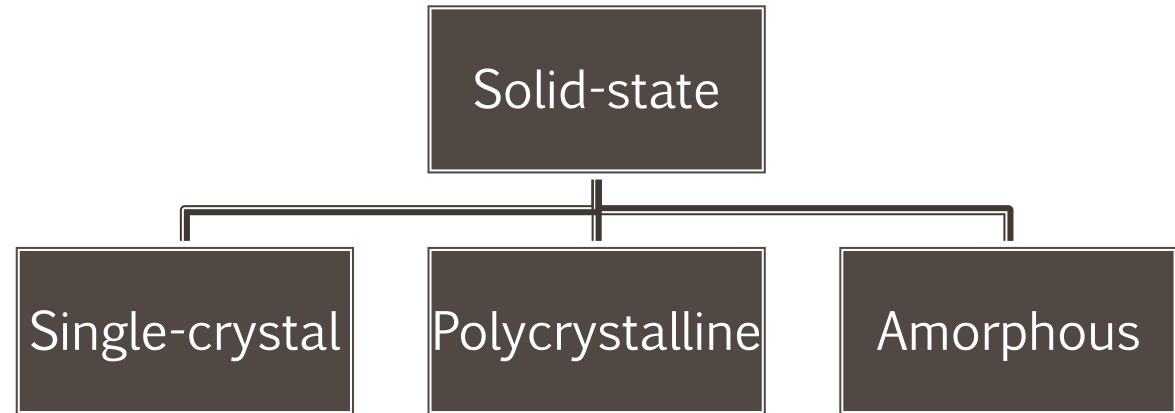
Table 1.2 | A list of some semiconductor materials

Elemental semiconductors	
Si	Silicon
Ge	Germanium
Compound semiconductors	
AlP	Aluminum phosphide
AlAs	Aluminum arsenide
GaP	Gallium phosphide
GaAs	Gallium arsenide
InP	Indium phosphide

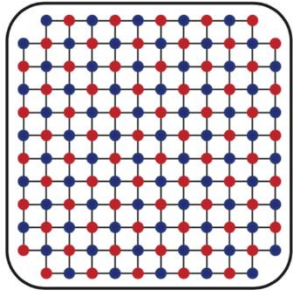
Solids and Crystals

Types of solids (1)

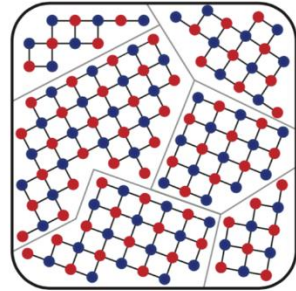
- Solid-state can be classified based on the **size of ordered region**
- **Ordered region:** atoms have regular geometric arrangement or periodicity



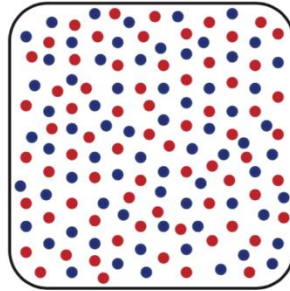
Types of solids (2)



Crystalline



Polycrystalline



Amorphous

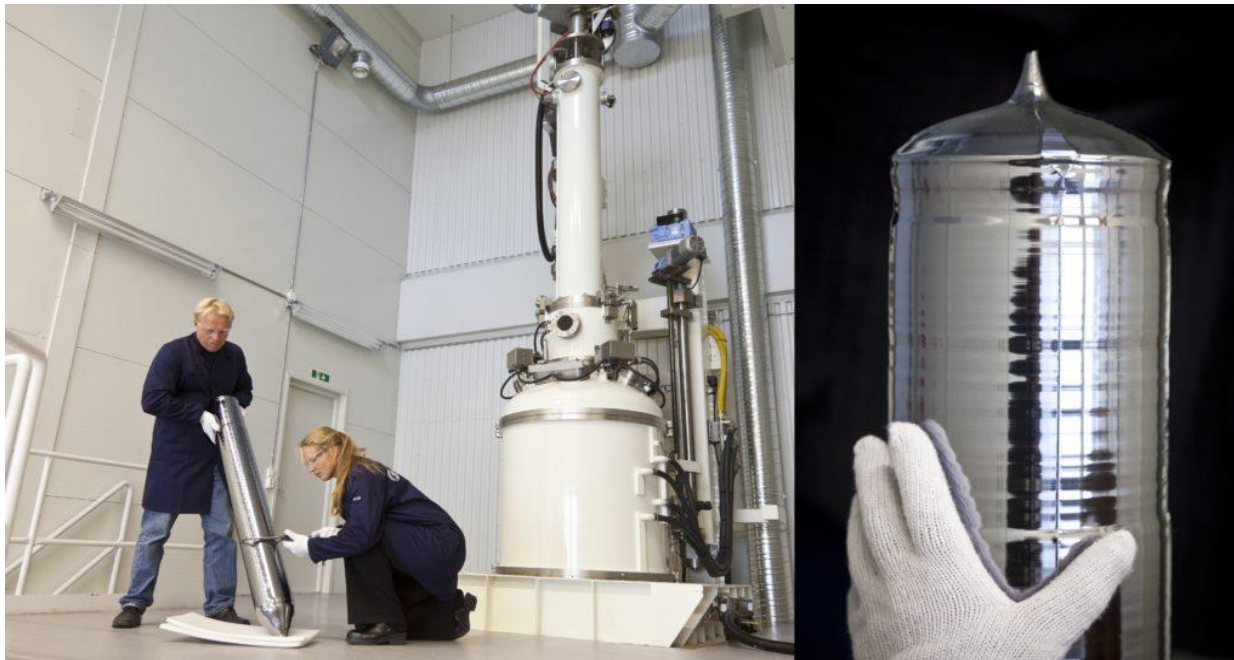
- **Single crystal:** order throughout the volume
- **Polycrystalline:** order through many atomic or molecular dimension
- **Amorphous:** only a few atomic or molecular dimension (**powdered form** of material)

- Property depends heavily on the arrangement (order) of atoms/molecules in a material

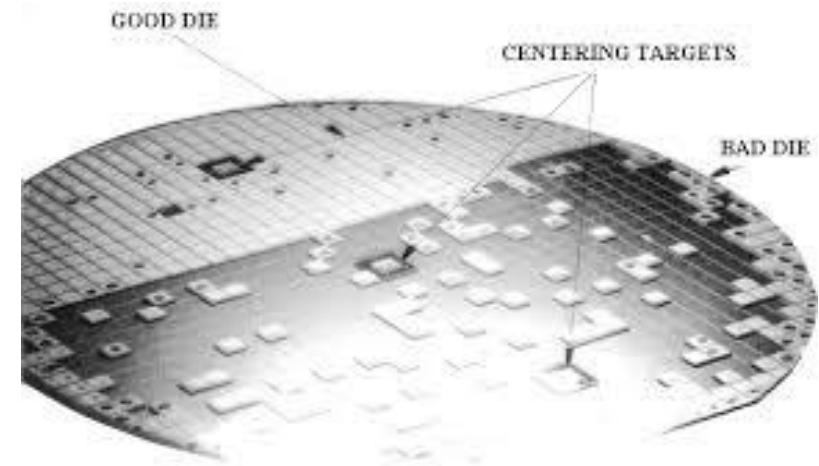
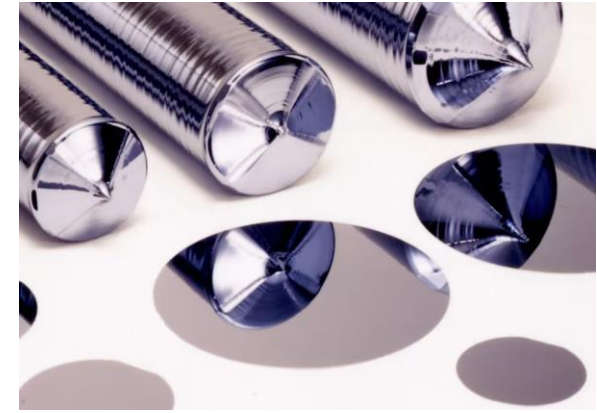
- Typically, single crystal of semiconductor material used for electronics:
 - Superior properties

Single crystal

- Silicon Ingots, Silicon Wafer, Dies: all are single crystal of silicon



Source: <https://susoltech.no/wp2-high-performance-silicon-ingots-and-wafers/>



Source: ST Microelectronics

Semiconductor Physics

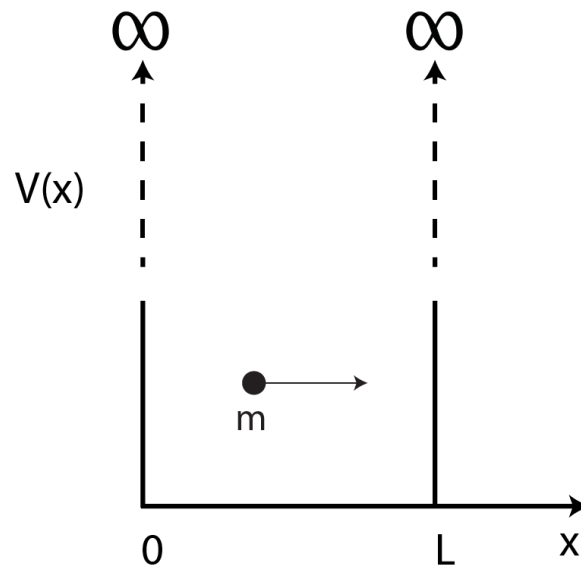
Preliminaries of Quantum Mechanics (1)

- To understand solid state devices, we need to understand the behavior of electrons under various conditions
 - Electron behavior is governed by Quantum Mechanics
 - Rigorous analysis of SSD requires concepts of Quantum Mechanics
 - We will try to understand SSD in this course by directly using some results/concepts of Quantum Mechanics (without rigorously deriving them)
-
- Wave-particle duality
 - Uncertainty principle
 - Quantum states
 - Pauli's Exclusion Principle

Quantization of Energy

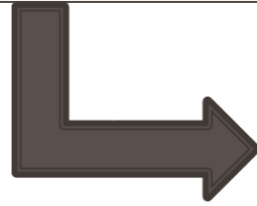
Important Result obtained using Quantum Mechanics

- Energy of bound particle is *quantized*



$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad \text{Energy is quantized}$$

Semiconductor Physics



Energy Bands

Formation of Energy Band (1)

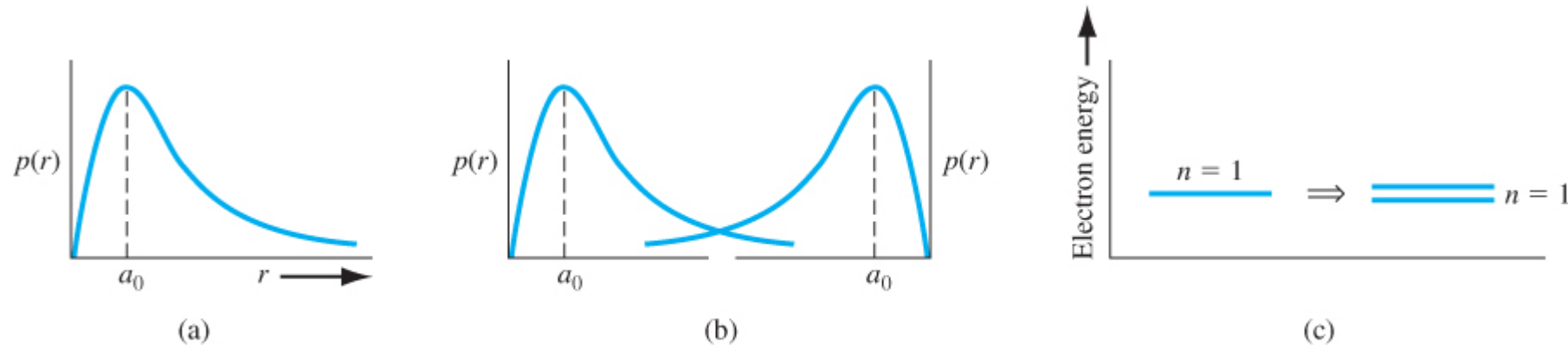


Figure 3.1 | (a) Probability density function of an isolated hydrogen atom. (b) Overlapping probability density functions of two adjacent hydrogen atoms. (c) The splitting of the $n = 1$ state.

- The probability of finding an electron $p(r)$ at a radial distance r can be obtained using the wave function Ψ (maximum at $a_0 = 0.529 \text{ \AA}$)
- When two hydrogen atoms are brought closer their wavefunctions interact
 - Electron energy split consistent with Pauli's exclusion principle

Formation of Energy Band (2)

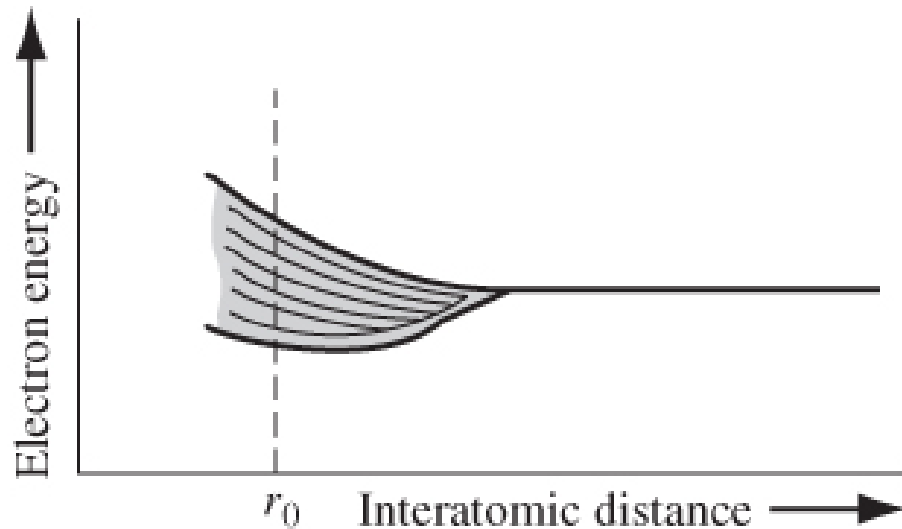


Figure 3.2 | The splitting of an energy state into a band of allowed energies.

Many atoms are brought in vicinity

- If many hydrogen-type atoms are brought in vicinity their wavefunctions will interact
- A distance r_0 will be taken at equilibrium
- The original energy level will split into several discrete energy levels to satisfy Pauli's exclusion principle

Formation of Energy Band (3)

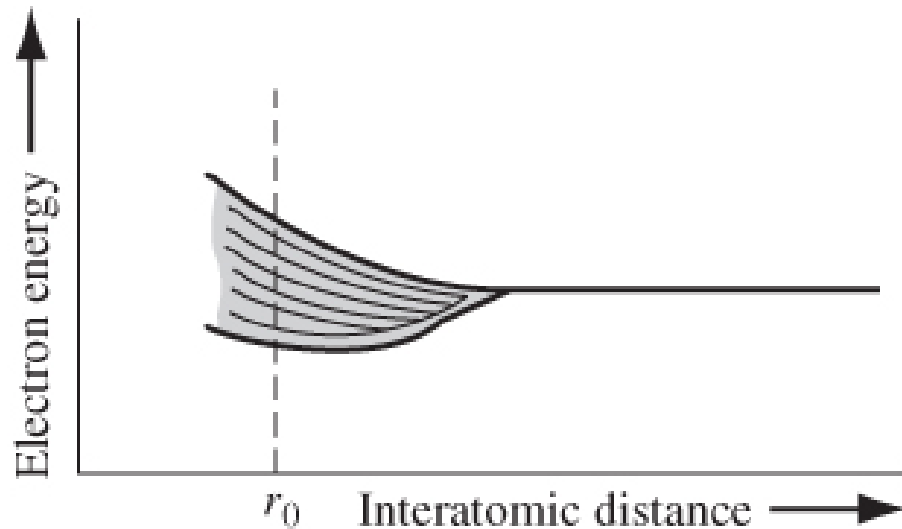


Figure 3.2 | The splitting of an energy state into a band of allowed energies.

Problem:

- Consider that the minimum and the maximum discrete energy levels are separated by 1 eV. Assume that there are 10^{19} electrons in the system and each occupy different energy states. What will be the spacing between the energy levels (assuming them to be uniformly separated)?

Ans: 10^{-19} eV

- Difference between discrete energy levels are so small that they can be considered as belonging to a “band” of energy level.
- Quasi-continuous* energy distribution

References

- D. A. Neaman, "*Semiconductor Physics and Devices*"